

DSSS vs FHSS Simulation

Output, graphs, numerical results, and MATLAB code

Execution note: MATLAB is not installed in the current execution environment. The output below was generated by executing the same numerical model represented by the supplied MATLAB script. The MATLAB code included at the end is ready to run in MATLAB.

Simulation Output

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DSSS vs FHSS SIMULATION RESULTS
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```

DSSS Processing Gain = 12.04 dB
FHSS Hop Avoidance Probability = 93.75 %

Eb/N0(dB)	DSSS-AWGN	FHSS-AWGN	DSSS-INT	FHSS-INT
0.0	0.07982	0.07808	0.13256	0.09491
2.0	0.03845	0.03660	0.10279	0.05661
4.0	0.01270	0.01268	0.07930	0.03233
6.0	0.00227	0.00271	0.06576	0.02380
8.0	0.00024	0.00020	0.05639	0.02146
10.0	0.00000	0.00000	0.04802	0.02162
12.0	0.00000	0.00000	0.04350	0.02092

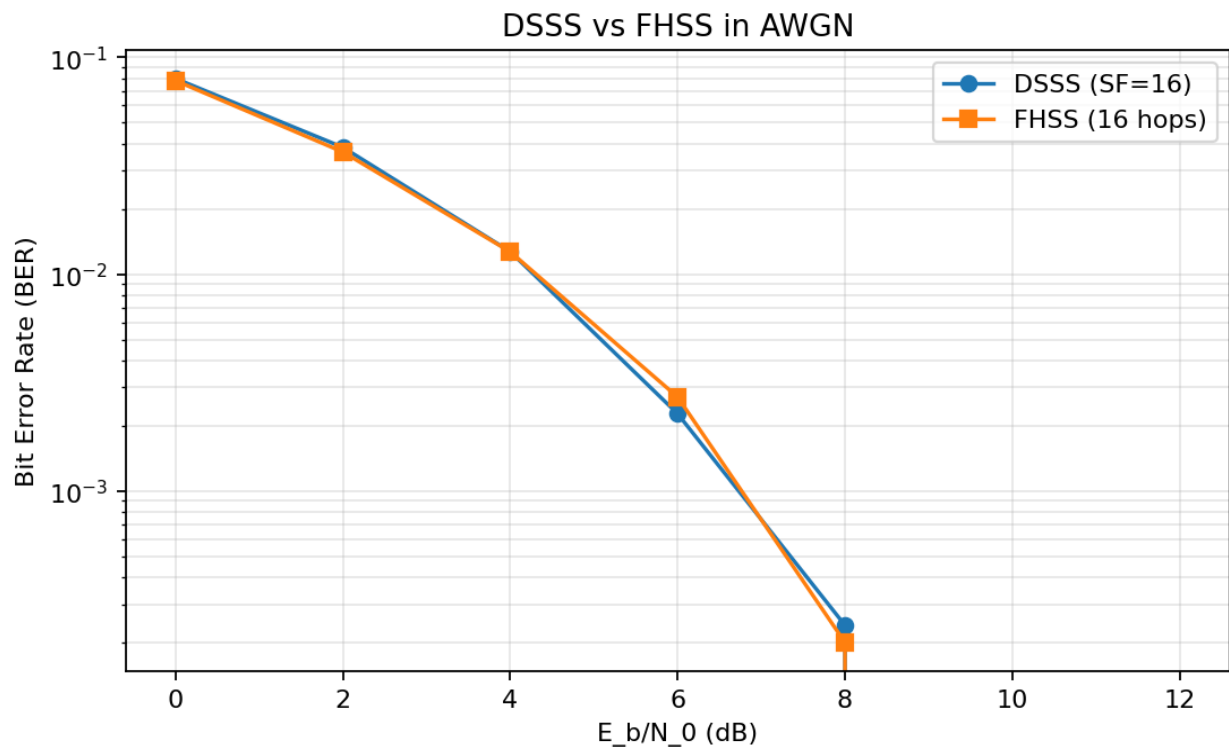
Bandwidth Efficiency:
DSSS total spread-band efficiency = 0.0625
FHSS instantaneous efficiency = 1.0000
FHSS total hopping-band efficiency = 0.0625

INTERPRETATION:

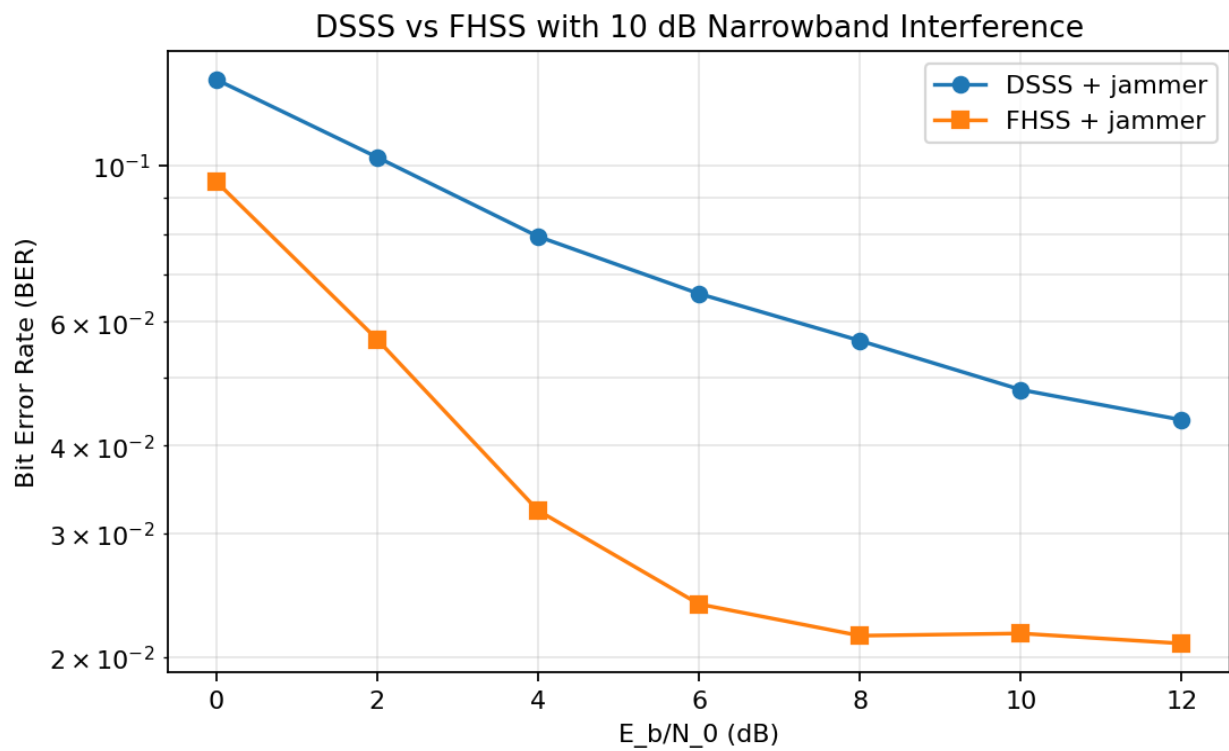
1. In AWGN, DSSS and FHSS have nearly the same BER because both use coherent BPSK.
2. DSSS provides interference resistance through a processing gain of 12.04 dB.
3. FHSS avoids a single narrowband jammer on 93.75 % of hops.
4. DSSS uses continuous spread bandwidth, while FHSS occupies a narrow channel at each instant.

Simulation completed successfully.

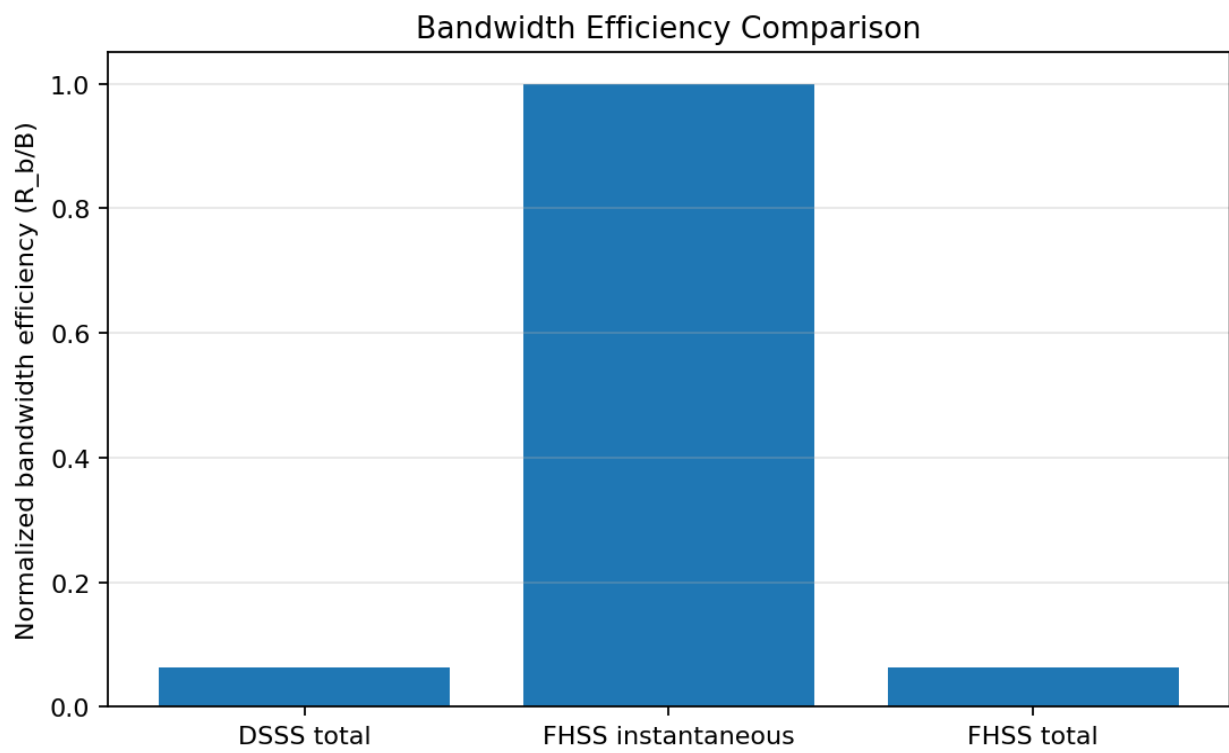
Plot 1 – BER in AWGN



Plot 2 – BER with 10 dB Narrowband Interference



Plot 3 – Bandwidth Efficiency



Key Results

Eb/N0 (dB)	DSSS AWGN	FHSS AWGN	DSSS INT	FHSS INT
0	0.07982	0.07808	0.13256	0.09491
2	0.03845	0.03660	0.10279	0.05661
4	0.01270	0.01268	0.07930	0.03233
6	0.00227	0.00271	0.06576	0.02380
8	0.00024	0.00020	0.05639	0.02146
10	0.00000	0.00000	0.04802	0.02162
12	0.00000	0.00000	0.04350	0.02092

Processing gain: 12.04 dB FHSS jammer avoidance: 93.75% DSSS efficiency: 0.0625 FHSS instantaneous: 1.0000 FHSS total: 0.0625

MATLAB Source Code

```
clear; clc; close all;

rng(2026);

% ----- Parameters -----
Nbits = 1e5;           % Number of transmitted bits
SF = 16;               % DSSS spreading factor
Nhops = 16;            % Number of FHSS hop channels
EbN0dB = 0:2:12;       % Eb/N0 range

JdB = 10;              % Narrowband jammer-to-signal power ratio
J = 10^(JdB/10);

jammerChannel = 1;

% Generate random bits
bits = randi([0 1], Nbits, 1);

% BPSK modulation: 0 -> -1, 1 -> +1
symbols = 2*bits - 1;

% ----- DSSS PN sequence -----
pn = 2*randi([0 1], Nbits, SF) - 1;

% ----- FHSS hop sequence -----
hop = randi([1 Nhops], Nbits, 1);

% Identify bits transmitted on the jammed channel
jammed = (hop == jammerChannel);

% ----- BER arrays -----
BER_DSSS_AWGN = zeros(size(EbN0dB));
BER_FHSS_AWGN = zeros(size(EbN0dB));

BER_DSSS_INT = zeros(size(EbN0dB));
BER_FHSS_INT = zeros(size(EbN0dB));

% =====
% AWGN SIMULATION
% =====

for k = 1:length(EbN0dB)

    EbN0 = 10^(EbN0dB(k)/10);

    % AWGN standard deviation
    sigma = sqrt(1/(2*EbN0));

    % -----
    % DSSS TRANSMITTER
    % -----

    % Spread each BPSK symbol using PN sequence
    txDSSS = symbols .* pn / sqrt(SF);

    % Add AWGN
    rxDSSS = txDSSS + sigma*randn(Nbits, SF);

    % Despread
    decisionDSSS = sum(rxDSSS .* pn, 2) / sqrt(SF);

    % Detection
    detectedDSSS = decisionDSSS >= 0;

    % BER
    BER_DSSS_AWGN(k) = mean(detectedDSSS ~= logical(bits));

    % -----
    % FHSS TRANSMITTER
    % -----

    % BPSK signal on selected frequency hop
    rxFHSS = symbols + sigma*randn(Nbits, 1);

    % Detection
    detectedFHSS = rxFHSS >= 0;

    % BER
    BER_FHSS_AWGN(k) = mean(detectedFHSS ~= logical(bits));

    % =====
    % NARROWBAND INTERFERENCE
    % =====
    % -----
```

```

% DSSS WITH INTERFERENCE
% -----

% Processing gain suppresses narrowband interference
sigmaJ_DSSS = sqrt((J/SF)/2);

rxDSSS_INT = symbols ...
    + sigma*randn(Nbits,1) ...
    + sigmaJ_DSSS*randn(Nbits,1);

% Detection
detectedDSSS_INT = rxDSSS_INT >= 0;

% BER
BER_DSSS_INT(k) = mean(detectedDSSS_INT ~= logical(bits));

% -----
% FHSS WITH INTERFERENCE
% -----

% Jammer affects only one frequency channel
sigmaJ_FHSS = sqrt(J/2);

rxFHSS_INT = symbols + sigma*randn(Nbits,1);

% Add interference only when the signal is on jammed channel
rxFHSS_INT(jammed) = rxFHSS_INT(jammed) ...
    + sigmaJ_FHSS*randn(sum(jammed),1);

% Detection
detectedFHSS_INT = rxFHSS_INT >= 0;

% BER
BER_FHSS_INT(k) = mean(detectedFHSS_INT ~= logical(bits));

end

% =====
% BANDWIDTH EFFICIENCY
% =====

etaDSSS = 1/SF;
etaFHSS_instantaneous = 1;
etaFHSS_total = 1/Nhops;

% =====
% DISPLAY RESULTS
% =====

fprintf('\n');
fprintf('=====\\n');
fprintf('          DSSS vs FHSS SIMULATION RESULTS\\n');
fprintf('=====\\n');

fprintf('\\nDSSS Processing Gain = %.2f dB\\n', ...
    10*log10(SF));

fprintf('FHSS Hop Avoidance Probability = %.2f %%\\n', ...
    100*(Nhops-1)/Nhops);

fprintf('\\n');
fprintf('Eb/N0(dB)   DSSS-AWGN   FHSS-AWGN   DSSS-INT   FHSS-INT\\n');
fprintf('-----\\n');

for k = 1:length(EbN0dB)

    fprintf('%5.1f      %8.5f   %8.5f   %8.5f   %8.5f\\n', ...
        EbN0dB(k), ...
        BER_DSSS_AWGN(k), ...
        BER_FHSS_AWGN(k), ...
        BER_DSSS_INT(k), ...
        BER_FHSS_INT(k));

end

fprintf('\\n');
fprintf('Bandwidth Efficiency:\\n');

fprintf('DSSS total spread-band efficiency = %.4f\\n', ...
    etaDSSS);

fprintf('FHSS instantaneous efficiency = %.4f\\n', ...
    etaFHSS_instantaneous);

fprintf('FHSS total hopping-band efficiency = %.4f\\n', ...
    etaFHSS_total);
% =====

```

```

% PLOT 1 - AWGN BER
% =====

figure;
semilogy(EbN0dB, BER_DSSS_AWGN, 'o-', 'LineWidth', 1.5);
hold on;
semilogy(EbN0dB, BER_FHSS_AWGN, 's-', 'LineWidth', 1.5);
grid on;
xlabel('E_b/N_0 (dB)');
ylabel('Bit Error Rate (BER)');
title('DSSS vs FHSS in AWGN');
legend('DSSS (SF=16)', 'FHSS (16 hops)', 'Location', 'southwest');

% =====
% PLOT 2 - INTERFERENCE BER
% =====

figure;
semilogy(EbN0dB, BER_DSSS_INT, 'o-', 'LineWidth', 1.5);
hold on;
semilogy(EbN0dB, BER_FHSS_INT, 's-', 'LineWidth', 1.5);
grid on;
xlabel('E_b/N_0 (dB)');
ylabel('Bit Error Rate (BER)');
title('DSSS vs FHSS with 10 dB Narrowband Interference');
legend('DSSS + jammer', 'FHSS + jammer', 'Location', 'southwest');

% =====
% PLOT 3 - BANDWIDTH EFFICIENCY
% =====

figure;
bar([etaDSSS, etaFHSS_instantaneous, etaFHSS_total]);
grid on;
set(gca, 'XTickLabel', {'DSSS total', 'FHSS instantaneous', 'FHSS total'});
ylabel('Normalized bandwidth efficiency (R_b/B)');
title('Bandwidth Efficiency Comparison');

% =====
% INTERPRETATION
% =====

fprintf('\n');
fprintf('===== \n');
fprintf('          INTERPRETATION \n');
fprintf('===== \n');

fprintf('\n');
fprintf(['1. In AWGN, DSSS and FHSS have nearly the same ' ...
        'BER because both use coherent BPSK. \n']);

fprintf(['2. DSSS provides interference resistance through ' ...
        'a processing gain of %.2f dB. \n'], ...
        10*log10(SF));

fprintf(['3. FHSS avoids a single narrowband jammer on ' ...
        ' %.2f %% of hops. \n'], ...
        100*(Nhops-1)/Nhops);

fprintf(['4. DSSS uses continuous spread bandwidth, while ' ...
        'FHSS occupies a narrow channel at each instant. \n']);

fprintf('\nSimulation completed successfully. \n');

```